

MEHMET FIRAT

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EDUCATION

Olin College of Engineering, Needham, MA

B.S. in Mechanical Engineering

Expected May 2028

Relevant Coursework: Statics; Thermal-Fluid Systems; Applied Math for Engineers; Engineering Systems Analysis; Mechanical Design

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, Onshape, FEA, DFM, thermal modeling, ergonomic design

Manufacturing: CNC machining, welding, 3D printing, drill press, band saw, angle grinder

Programming: Python, MATLAB, Java

RELEVANT EXPERIENCE

Cibo Vita

Jun 2026 - Aug 2026

Manufacturing Engineering / Operations Intern

- Developed a scraper for a guillotine-style cutter, eliminating recurring product blockages and **roughly 2-minute cleaning stoppages occurring about once per hour**.
- Designed a modular cleaning cart with custom holders for **120 MULTIPOND components** to organize part handling during sanitation.
- Iterated hopper-holder geometry in CAD and validated fit through 3D-printed prototypes, evaluating manufacturability, material use, and operator loading.
- Observed low-output production equipment to identify causes of reduced throughput; presented mechanical and process-improvement recommendations.

Olin College Robotics Lab (Farm Robotics)

Sep 2024 - Present

Mechanical Design Team Member | Co-Mechanical Design Team Lead since Fall 2026

- Contributed to HydroFleet, winner of the **2026 Farm Robotics Challenge Excellence in Drone Applications Award (Division I)**.
- Designed and tested **2 drill-style soil-probe prototypes** for drone deployment under a **700 g** payload requirement; demonstrated **6 in** soil insertion in prototype testing.
- Iterated lead-screw, auger, and anchoring assemblies using bench and soil tests to address slippage, alignment, and stability issues.
- Supported GPS implementation for an autonomous weeding robot's field navigation.

Olin College Formula Team

Sep 2024 - May 2026

Cockpit / Drivetrain

- Designed and fabricated dashboard and "Nostril" enclosure in SolidWorks, using ergonomic modeling and FEA to reduce mass by **approximately 20%**.
- Supported drivetrain design reviews and integration checks; evaluated manufacturability, tolerances, and mechanical interfaces to reduce assembly risk.
- Integrated heat sinks and airflow features; coordinated packaging and mechanical interfaces with electrical and aerodynamics subteams.

Boston Education & Counseling

Oct 2025 - Present

Robotics Mentor

- Mentor students through robot builds, troubleshooting mechanical design, wiring, and fabrication.